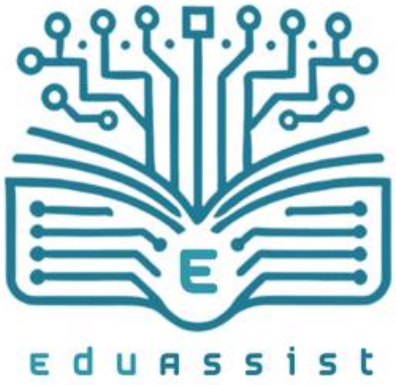
Imam Abdulrahman Bin Faisal University
College of Computer Science & Information Technology
Department of Computer Science

CS 411 – Software Engineering
Term 1 – 2024/2025

Software Project Management Plans (SPMP)

For



Version [01]

Students:

[Advisor]

Supervised By:
Dr. Enas Elsharawy

This Software Project Management Plans (SPMP) was prepared and provided as a deliverable for Software Engineering,CS411.Term 1, and it will be used by EDUassist.This document is based in part on the IEEE Recommended Practice for SPMP Descriptions.

Team Members

#	Name	ID	Role
1			Leader
2			Member
3			Member
4			Member

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Revision History

Name	Date	Reason For Changes	Version
All members	Sep 30, 2024	Prepared initial version	0.1
All members	Oct. 1, 2024	Updated section 1	0.2
All members	Oct. 2-3, 2024	Updated section 2,3	0.3
All members	Oct. 4-6, 2024	Updated section 4,5,6	0.4
All members	Oct. 7, 2024	Updated section 3	0.5
All members	Oct. 8, 2024	Updated section 4,5	0.6
All members	Oct. 9, 2024	Spelling and grammar mistake	0.7
All members	Oct. 12, 2024	Complete review - Final version	1.0

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Phases. Adapted from [15]

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1. Project Overview

This section of the provides an overview of the purpose, scope, and objectives of the project for which the Plan has been written, the project assumptions and constraints, a list of project deliverables, a summary of the project schedule and budget.

1.1 Purpose, Scope, and Objectives

The EDUassist project aims to tackle the challenges students face in managing their academic responsibilities and enhancing their learning experiences. In a rapidly evolving educational landscape, traditional study planning tools often fail to meet the diverse and personalized needs of learners. EDUassist seeks to provide a comprehensive, AI-driven solution that integrates various academic tools into a cohesive platform, improving organization, collaboration, and overall academic performance.

Users can log in via email or phone number, secured by a strong password and a verification code. Once logged in, students can access personalized study plans, collaborative tools, and AI-powered resources tailored to their needs. The platform offers several key features designed to enhance the academic journey of its users.

EDUassist provides various features specifically tailored for students. All users must comply with the EDUassist terms of use; violations may result in penalties, including restricted access to certain functionalities.

The scope of the EDUassist project utilizes a Plan-driven Waterfall model, allowing for a linear and sequential approach tailored for students. Key features will include:

Service Features	Description	Estimated Price
Study Plan Organization	Automatically interprets syllabus data from PDFs into a user-friendly weekly schedule. Supports time management and progress tracking for students.	Free
App Integration	Seamless integration with platforms like Google Calendar to synchronize academic schedules and tasks.	Free
Collaborative Whiteboard Functionality	Offers an interactive shared whiteboard for group brainstorming, idea mapping, and includes AI-powered support for content suggestions and real-time error correction.	99 - 200 SR
Visual Learning Tools	Converts lecture materials and text into animated visual content (e.g., cartoons, diagrams) to enhance understanding. Supports interactive visuals for diverse learning styles.	239 SR (for one person)
Resource Recommendations	AI algorithms analyze individual academic needs to recommend personalized study materials, such as videos, articles, and books relevant to coursework.	Free
Peer Collaboration Platform	Features for students to connect and collaborate, enabling knowledge sharing through tools like chat and video calls.	Free

Table 2 : Services Features

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Software Objectives:

- Accurately interpret syllabus data and create personalized weekly schedules.
- Seamlessly integrate with external platforms for efficient task management.
- Offer collaborative features for enhanced group work and brainstorming.
- Provide AI-powered visual aids to boost understanding and retention.
- Recommend personalized study materials based on individual learning preferences.

Additional Objectives:

- Improve organization and time management skills.
- Foster collaboration and teamwork among students.
- Enhance user engagement through interactive features.
- Promote personalized learning experiences.
- Develop critical thinking and problem-solving abilities.
- Support continuous feedback and improvement in academic performance.
- Encourage the use of technology to facilitate learning.
- Support files in Arabic.



Figure 1 : Overall Objectives

1.2 Assumptions, Constraints and Risks

This section outlines the assumptions, constraints, and risks that must be considered for the successful implementation of the project. The key points are as follows:

Assumptions:

- Resources will be available as required.
- The budget for the project will be adhered to.
- All risks that may cause delays in project delivery will be evaluated.
- Customer requirements will be feasible and align with the team's capabilities.

Constraints:

- The project must remain within the allocated budget.
- The project should realistically reflect the intended scenery.
- Work will be conducted with the available resources.
- The project must be completed within the specified delivery timeline.
- The core idea of the proposal must be maintained.

Risks:

- The use of low-quality software may hinder the creation of a realistic environment.
- New technologies may require additional experience and skills from team members.
- The delivered project may not meet expected quality standards.
- The project may take longer than scheduled.
- The budget may be exceeded.

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1.3 Project Deliverables

The EDUassist project will produce a series of official documents designed to assist students. The software documentation and related deliverables, shown in **Table 3** , will be provided in softcopy format unless specified otherwise. All deliverables will be made available online, with each deliverable provided in a single copy unless additional versions are required.

Deliverable	Date	Delivery Method
Project Proposal	Sep 30, 2024	Softcopy
Software Project Management Plan (SPMP)	Oct. 1, 2024	Softcopy
Software Requirements Specification (SRS)	Oct. 2-3, 2024	Softcopy
Software Design Specification (SDS)	Oct. 4-6, 2024	Softcopy
Software Test Plan (STP)	Oct. 7, 2024	Softcopy
Final Report	Oct. 8, 2024	Softcopy

Table 3: Project Deliverables

1.3.1 Delivery Media

- All deliverables will be submitted as PDF documents unless otherwise specified.
- Documents larger than 50MB will be compressed and submitted as ZIP files.

1.3.2 Special Instructions for Packaging and Handling

- File Naming Convention: All files must follow the naming convention: EDUassist_[Deliverable]_[VersionNumber].pdf (e.g., EDUassist_SRS_v1.0.pdf).
- Backup: All deliverables must be backed up on the local system before submission to avoid data loss.

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1.4 Schedule and Budget Summary

The project will be completed over a period of 11 to 12 months with a total budget of 150,000 Saudi Riyal (SR). The budget and timeline are allocated across different phases of the project, each focusing on key activities required to successfully deliver the final product.

Workflow	Duration	Team Members	Budget (SR)
Requirement Gathering	4 weeks	2 members	35,000
Analysis	4 weeks	Analysis Group (2 members)	7,500
Design	11 weeks	Analysis Group (2 members)	25,000
Implementation	22 weeks	Programming Group (2 members)	70,000
Testing	9 weeks	3 team members	10,000
Deployment and Submission	1 week	-	-

Table 4: Summary of the Budget

1.5 Evolution of the Plan

This project plan adheres to IEEE standards and follows a plan-driven approach. Regular evaluations of the project documentation will be conducted to ensure compliance with the established plan and to track progress against defined milestones.
Team members will collaborate to discuss and implement changes as needed to align with project objectives and enhance overall results.

1.6 References

1. Title: A SPMP Changeability Measurement Model for Reducing Software Development Risk
Report Number: Unspecified
Date: 2012
Author: Unspecified
Publishing Organization: SpringerLink
Link: SpringerLink

2. Title: IEEE Standard for Software Project Management Plans
Report Number: IEEE 1058-1998
Date: 1998
Author: IEEE Standards Association
Publishing Organization: IEEE
Link: SpringerLink

3. Title: The Impact of Organizational Structure on Project Performance in the Energy Sector
Report Number: Unspecified
Date: Unspecified
Author: Unspecified
Publishing Organization: SpringerLink
Link: SpringerLink

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4. Title: Project Organization
Report Number: Unspecified
Date: 2022
Author: Unspecified
Publishing Organization: Springer
Link: SpringerLink

5. Title: Managerial Processes: Business Process That Sustain Performance
Report Number: Unspecified
Date: 2011
Author: Unspecified
Publishing Organization: Emerald Insight
Link: Emerald Insight

6. Title: Business Process Performance Measurement: A Structured Literature Review
Report Number: Unspecified
Date: Unspecified
Author: Unspecified
Publishing Organization: Springer Open
Link: Springer Open

7. Title: Software Project Planning with Tracking and Oversight
Report Number: Unspecified
Date: 2017
Author: Unspecified
Publishing Organization: International Journal of Scientific and Research Publications (IJSRP)
Link: IJSRP

8. Title: Project Tracker: The Ultimate Guide to Project Tracking
Report Number: Unspecified
Date: Unspecified
Author: Unspecified
Publishing Organization: ProjectManager.com
Link: ProjectManager.com

9. Title: Towards Successful Business Process Improvement
Report Number: Unspecified
Date: Unspecified
Author: Unspecified
Publishing Organization: PLOS One
Link: PLOS One

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10. Title: 8 Steps to Write an Impressive Technical Approach
Report Number: Unspecified
Date: Unspecified
Author: Unspecified
Publishing Organization: Meridian West
Link: Meridian West

11. Title: Software Engineering Methodologies: A Review of the Waterfall Model and Object-Oriented Approach
Report Number: Unspecified
Date: 2013
Author: Adenowo, A. A., & Adenowo, B. A.
Publishing Organization: International Journal of Scientific & Engineering Research

12. Title: Waterfall Methodology, Prototyping and Agile Development
Report Number: Unspecified
Date: 2021
Author: Senarath, U. S.
Publishing Organization: Tech Report

13. Title: Imam Abdulrahman Bin Faisal University - CS411 Course Material
Report Number: EDUassist Project SPMP
Date: Unspecified
Author: Unspecified (prepared by students under the supervision of Dr. Enas Elsharawy)
Publishing Organization: Imam Abdulrahman Bin Faisal University

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1.7 Definitions and Acronyms

All the acronyms that are used in this document are listed in **Table 5** below

Acronyms	Stand for	Definition
SPMP	Software Project Management Plans	A formal document that describes how a project should be implemented effectively.
SRS	Software Requirements Specification	A comprehensive document that specifies the functional and non-functional requirements of the software, ensuring alignment with stakeholder expectations and technical feasibility.
SDS	Software Design Specification	A detailed technical document that defines the architectural structure, component interactions, and overall design of the software system.
STP	Software Test Plan	A formal document outlining the testing strategy, objectives, scope, resources, and schedule for testing activities to ensure the software meets its quality standards
HA	Hybrid Approach	A combination of methodologies that uses existing tools alongside custom development for enhanced efficiency and tailored solutions.
V&V	Verification and Validation	The dual processes of confirming that the software product meets its specified requirements (verification) and that it fulfills its intended use in the operational environment (validation).
WBS	Work Breakdown Structure	A project management tool that provides a hierarchical decomposition of the project scope into manageable work packages to facilitate effective planning, resource allocation, and monitoring.
CCB	Change Control Board	A formally designated group responsible for reviewing, evaluating, and approving or rejecting proposed changes to the project scope, schedule, or deliverables.
GUI	Graphical User Interface	The interactive interface of a software system through which users engage with the system's functionality via visual elements such as buttons, menus, and icons.
UML	Unified Modeling Language	A standardized modeling language used to specify, visualize, construct, and document the components of a software system’s design, particularly for object-oriented development.
AWS	Amazon Web Services	A cloud computing platform offering a range of services including computing power, storage, and databases, designed to provide scalable and secure infrastructure solutions for businesses.
ER Diagram	Entity-Relationship Diagram	A graphical representation used in database design that illustrates the entities within a system and the relationships between them

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IEEE	Institute of Electrical and Electronics Engineers	A professional organization that develops industry standards for technology and software engineering, promoting best practices in various fields of electrical, electronics, and computing.
AI	Artificial Intelligence	The development of computer systems capable of performing tasks that normally require human intelligence, such as visual perception, decision-making, and language understanding.
SQL	Structured Query Language	A standardized language used for managing, querying, and manipulating relational databases, essential for backend data management in software applications
JUnit	Java Unit	A widely-used testing framework for Java programming language, enabling the execution of repeatable automated tests to ensure code quality and functionality.
EVM	Earned Value Management	A project management methodology that integrates scope, schedule, and cost metrics to assess project performance and progress against a baseline.
UX/UI	User Experience / User Interface	UX refers to the overall experience a user has while interacting with a system, focusing on ease of use and satisfaction. UI refers to the visual layout and interaction design of the system's interface.
PDF	Portable Document Format	A widely-used file format that preserves the layout, fonts, images, and graphics of documents across different devices and operating systems.

Table 5: Definitions and Acronyms

1.8 Document Structure

1. Project Overview

This section introduces the project's purpose, scope, and objectives, outlining how our software solution addresses the identified problem. We will define the key deliverables, summarize the schedule and budget, and detail assumptions, constraints, and risks. This overview sets the foundation for project expectations and goals, ensuring alignment with stakeholders.

2. Project Organization

We describe the structure of our team and how we interface with external systems and stakeholders. The External Interfaces section outlines interactions with third-party platforms, while the Internal Structure covers the roles and responsibilities within our team. This ensures clarity in communication and the division of tasks across the project lifecycle.

3. Managerial Process Plans

This section outlines how we will manage the project from start to finish. The Start-up Plan provides estimates for time, budget, and resources. The Work Plan includes a detailed Work Breakdown Structure (WBS), resource allocation, and schedule. We will use the Tracking Plan to monitor progress, and the Risk Management Plan to mitigate any identified risks. Finally, the Project Closeout Plan describes how we will complete and review the project.

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4. Technical Process Plans

We detail the technical aspects of the project here. The Process Model explains the development approach (e.g., Waterfall), while the Methods, Tools, and Techniques section lists the technologies we will use. The Infrastructure describes the required technical environment, and the Product Acceptance outlines the criteria for delivering a successful product.

5. Supporting Process Plans

This section covers essential support activities, particularly in relation to project documentation and quality assurance. We explain how we will produce, maintain, and manage documentation throughout the project. This section also outlines the processes that ensure project quality is upheld.

6. Additional Plans

Here, we present other plans critical to project success, such as the Maintenance Plan for post-deployment support and the Deployment Plan detailing how the software will be released. These plans ensure long-term project sustainability and user satisfaction.

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2. Project Organization

2.1 External Interfaces

The EDUassist project interfaces with external entities to ensure that the software meets the requirements and expectations of its users. The primary interaction occurs between the EDUassist platform and the external users, while the development team works in the background to ensure smooth functionality and timely updates. The EDUassist platform serves as the main interface for external entities, particularly the students. All interactions, feedback, and usage flow through the platform, enabling students to engage with the system for academic management, collaboration, and support. The students, as the primary external users, directly interact with the platform to manage their academic tasks and schedules. Their feedback, queries, and suggestions are gathered through the EDUassist system, which acts as the communication bridge between the users and the development team. Meanwhile, the EDUassist development team operates behind the scenes, maintaining and enhancing the platform based on the input received from the students. The team ensures that the platform runs smoothly and evolves in line with user needs.

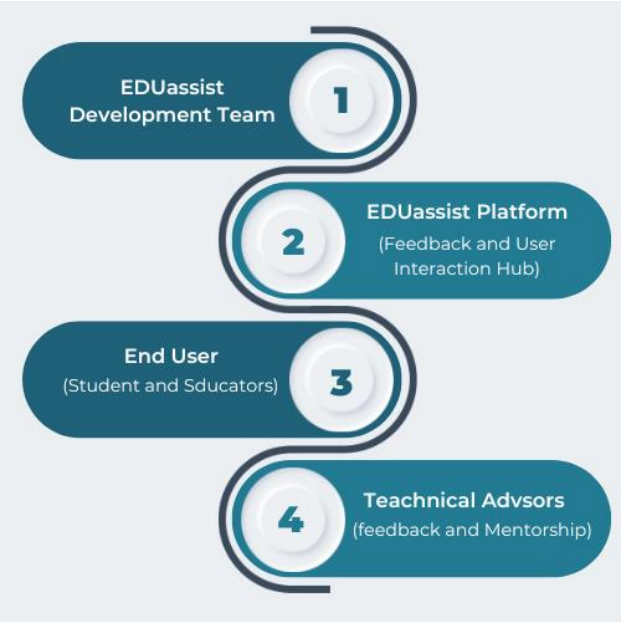


Figure 2: External Interfaces

2.2 Internal Structure

The EduAssist project will be developed collaboratively by , , , and under the supervision of Dr. Enas El-Sharawi. The team will convene for weekly in-person meetings to review project progress, supplemented by online sessions via Zoom to facilitate ongoing communication.

The internal structure of the team comprises individuals with diverse expertise, ensuring a well-rounded approach to project development. Each member will leverage their skills in areas such as technical writing, design, and project management to contribute effectively.

Figure 3 presents a hierarchical representation of the team structure, while Figure 4 outlines the organizational structure, delineating the lines of authority, responsibility, and communication pathways

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within the project.

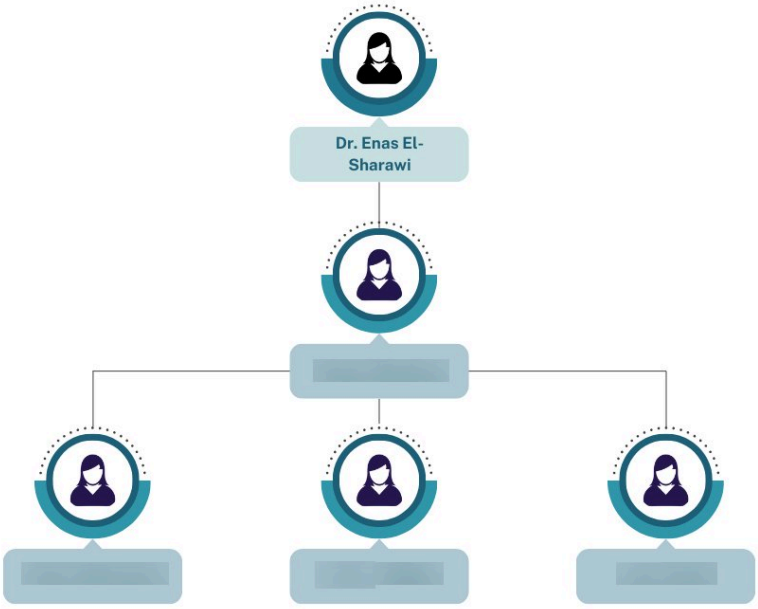


Figure 3: Internal Structure

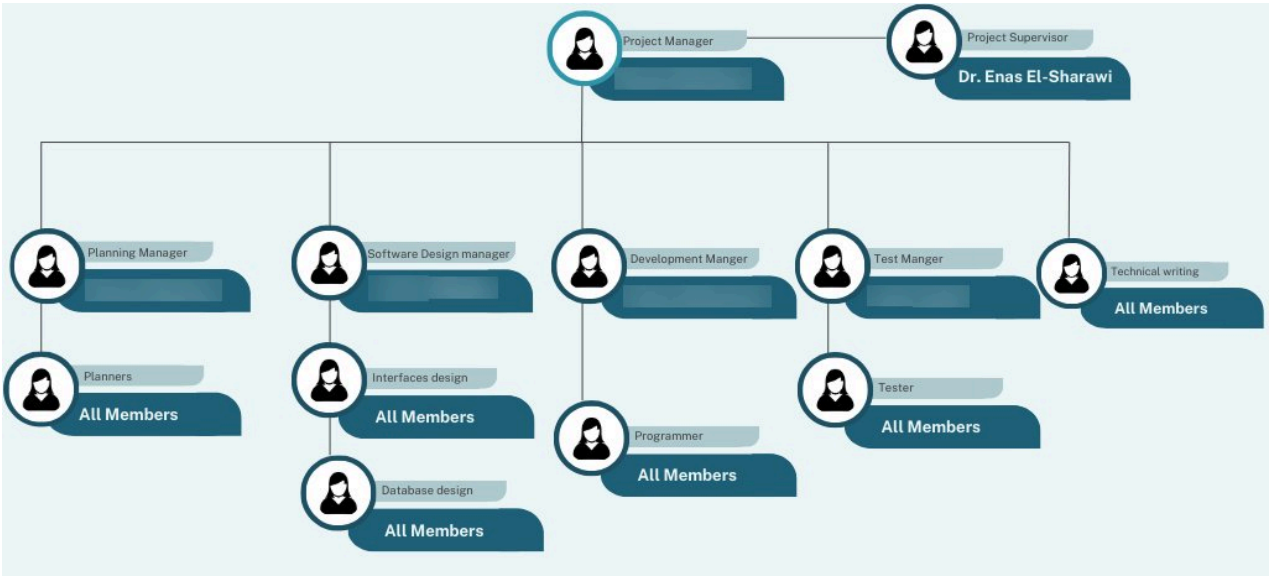


Figure 4 : :Organizational Structure

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2.3 Roles and Responsibilities

It is essential to clearly define the roles and responsibilities of each team member in the EduAssist project based on their individual skills and experiences. The project manager, utilizing their knowledge of the team's strengths, will determine the allocation of tasks and responsibilities, as illustrated in **Figure 5.** Moreover, various tools will support collaboration and communication among team members. For example, Microsoft Word can be used for documentation, while Canva and Figma may assist in design tasks, and project planning can be organized through Microsoft Planner. This approach ensures that each member contributes effectively to the project's success.

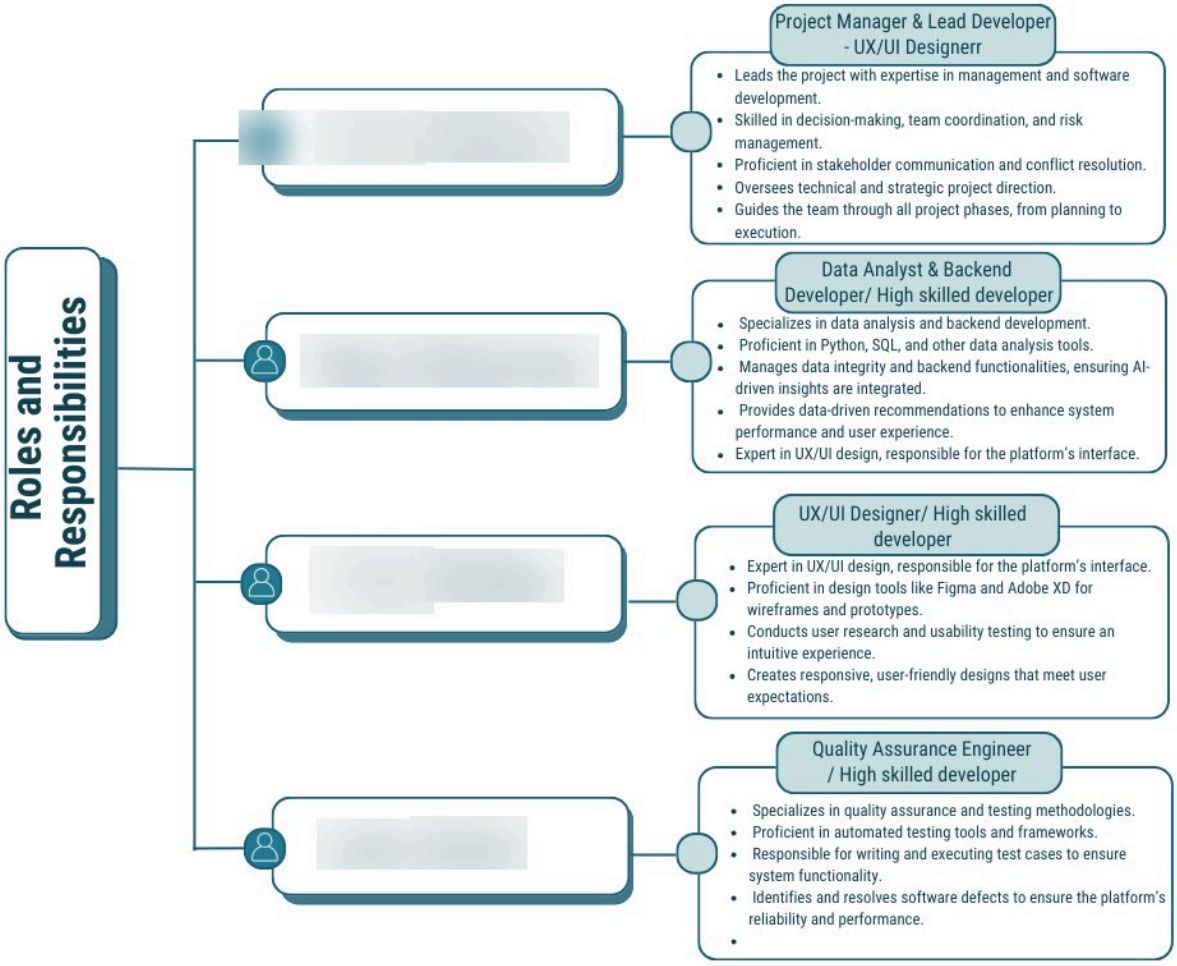


Figure 5 : Roles and responsibilities

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3. Managerial Process Plans

This section of the Project Management Plan specifies the project management processes for the project. This section defines the plans for project start-up, risk management, project work, project tracking and project close-out.

3.1 Start-up Plan

3.1.1 Estimates

The project estimation for EDUassist involves a comprehensive assessment of cost, schedule, and resource requirements, each accompanied by associated confidence levels. The estimated cost has been projected at approximately $150,000, with a confidence level of 80%, based on historical data from similar projects. The schedule outlines a timeline of 11-12 months for completion, also with a confidence level of 85%, derived from previous project experience and expert judgment. Resource requirements include a team of four members, with a projected allocation of 600 hours of total work, supported by a confidence level of 70%.

To estimate these parameters, we employed several methods, including analogy-based estimation, where we referenced past projects with similar scopes; expert judgment, which incorporated insights from team members experienced in software development; and standard unit sizes that align with industry benchmarks. The primary sources of our estimate data include a historical database of prior software engineering projects and cost models specific to educational technology.

Re-estimation will be conducted using techniques such as earned value management (EVM) to track project performance against the baseline and provide insights for adjustments. Additionally, we will utilize software tools like Microsoft Project for scheduling re-estimations and resource allocation assessments. The schedule for re-estimation will be event-driven, occurring at key project milestones (e.g., completion of the Software Requirements Specification and Software Design Specification) as well as on a regular monthly basis to ensure that we remain aligned with project goals and timelines. This structured approach will allow for timely adjustments and enhance the accuracy of our forecasts throughout the project lifecycle.

3.1.2 Staffing

The EDUassist project will be executed by a highly skilled team of four members, each bringing critical expertise. The team’s roles and responsibilities are clearly defined to ensure the successful development and deployment of the platform. Each member's role is outlined below:

- (Project Manager & Lead Develop)

  - Leads the project with expertise in project management and software development.

  - Skilled in decision-making, team coordination, and risk management to ensure the project stays on track.

  - Proficient in stakeholder communication and conflict resolution.

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- Oversees the technical and strategic direction of the project.
- Guides the team through the project lifecycle, from planning to execution.

- (Data Analyst & Backend Developer/ High skilled developer)
  - Specializes in data analysis and backend development.
  - Proficient in Python, SQL, and other data analysis tools.
  - Manages data integrity and backend functionalities, ensuring AI-driven insights are integrated.
  - Provides data-driven recommendations to enhance system performance and user experience.

- (UX/UI Designer/ High skilled developer)
  - Expert in UX/UI design and responsible for the platform's user interface.
  - Proficient in design tools like Figma and Adobe XD for wireframes and prototypes.
  - Conducts user research and usability testing to ensure an intuitive user experience.
  - Creates responsive, user-friendly designs that meet user expectations.

- (Quality Assurance Engineer/ High skilled developer)
  - Specializes in quality assurance and testing methodologies.
  - Proficient in automated testing tools and frameworks.
  - Responsible for writing and executing test cases to ensure system functionality.
  - Identifies and resolves software defects to ensure the platform's reliability and performance.

Team Structure and Synergy

The combined expertise of the team ensures that all critical aspects of the project are handled efficiently. 's leadership guarantees smooth project coordination and risk management, while 's backend and data analysis skills ensure a solid, data-driven foundation for the platform. 's focus on design ensures an intuitive and appealing user experience, and

's dedication to quality assurance ensures the system is thoroughly tested and ready for deployment. This balanced distribution of responsibilities allows the team to effectively collaborate and tackle each phase of the project with precision.

The various roles and skills required for the EDUassist project team are summarized in **Table 6**, along with the specific competencies associated with each position. This table serves as a valuable resource for understanding the expertise needed to execute the project effectively. By presenting this information in a clear tabular format,

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stakeholders and team members can easily grasp the overall capabilities of the team, facilitating better planning and coordination of their activities

#	Project Phases	Human Resource	Duration
1	Project Proposal	Team Members	One Week
2	Project Management Plan		Two Weeks
3	Project Requirements Specification (SRS)		Two Weeks
4	Project Design (SDS)	Team Members Project Supervisor and Client	Two Weeks
5	Project Test Plan (STP)		Two Weeks
6	Final Project		One Week

Table 6 : Project phases

3.1.3 Project Staff Training

For the project's success, all team members must have a combination of managerial, technical writing, design, and programming skills. Each member will need management abilities in project planning, design, development, or testing, while also being proficient in technical writing to document project requirements and progress. These skills will ensure efficient task execution, collaboration, and clear communication. **Table 7** outlines the training, focusing on technical and soft skills to improve team performance.

Skill Type	Skill Needed	Training Method	Personal
Managerial	-Leadership and decision-making skills -Communication skills -Emotional intelligence	-Offering manuals and guides for project management. -Solving exercises	All
Technical writing	-Writing tools -Understanding simplification -Research skills	-Online courses -Workshops	All
Programming	-Problem-Solving -Data Structures and Algorithms -Testing	-Coding Challenges -Peer Programming -Regular Practice	All
Design	-Visual Design -ux/ui design -collaboration and Communication	-Online Courses -Workshops and Seminars	All

Table 7 : Training skills

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3.2 Work Plan

3.2.1 Work Breakdown Structure

The Work Breakdown Structure (WBS) is vital for effective risk management, scheduling, cost estimation, and project planning. It outlines the work activities for the EduAssist project and their interrelationships. Each task will be broken down to identify risk factors and facilitate precise estimation of resources and durations. **Table 8** details the WBS, specifying activities, estimated durations, necessary resources, and deliverables, while **Table 9** will define acceptance criteria and related predecessor and successor activities.

	Work Activities	Sub Work Activities	Estimated Duration	Status	Delivery Method
1	Project Definition	Brainstorming	4 Days	Completed	Softcopy
2	Project Proposal	- Preparing Project proposal - Project logo	6 Days	Completed	-
2.1	Submitting Project Proposal	-	1 Days	Completed	Softcopy
3	Project Management Plan (SPMP)	- Project Overview - Project Organization - Managerial Process Plans - Technical Process Plans - Supporting Process Plans - Additional Plans	10 Days	In progress	-
3.1	Submitting SPMP	Review	2 Days	In progress	Softcopy
4	Project Requirement	- Collect Requirements - Analyzing the Requirement - Preparing Project Requirement (SRS)	13 Days	Incomplete	-
4.1	Submitting Project Requirement	Review	1 Days	Incomplete	Softcopy
5	Project Interface Design (SDS)	- Project Color Scheme	14 Days	Incomplete	Softcopy
6	Project Test Plan (STS)	-	14 Days	Incomplete	Softcopy
6.1	Build the Database	-	7 Days	Incomplete	-
6.2	Performance testing	-	7 Days	Incomplete	-
7	Project Presentation	-	1 Days	Incomplete	-

Table 8: Work Breakdown Structure

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3.2.2 Schedule Allocation

Table 8 outlines the time allocated for each task in the EduAssist project, as illustrated in **Figure 6,** which utilizes Microsoft Planner to cover the timeline from the start of week 1 to the end of week 13. The duration required to complete each task ranges from one to three weeks. The team leader conducts weekly oversight and evaluation of the schedule to ensure that all team members are meeting their deadlines and that the project progresses as intended.

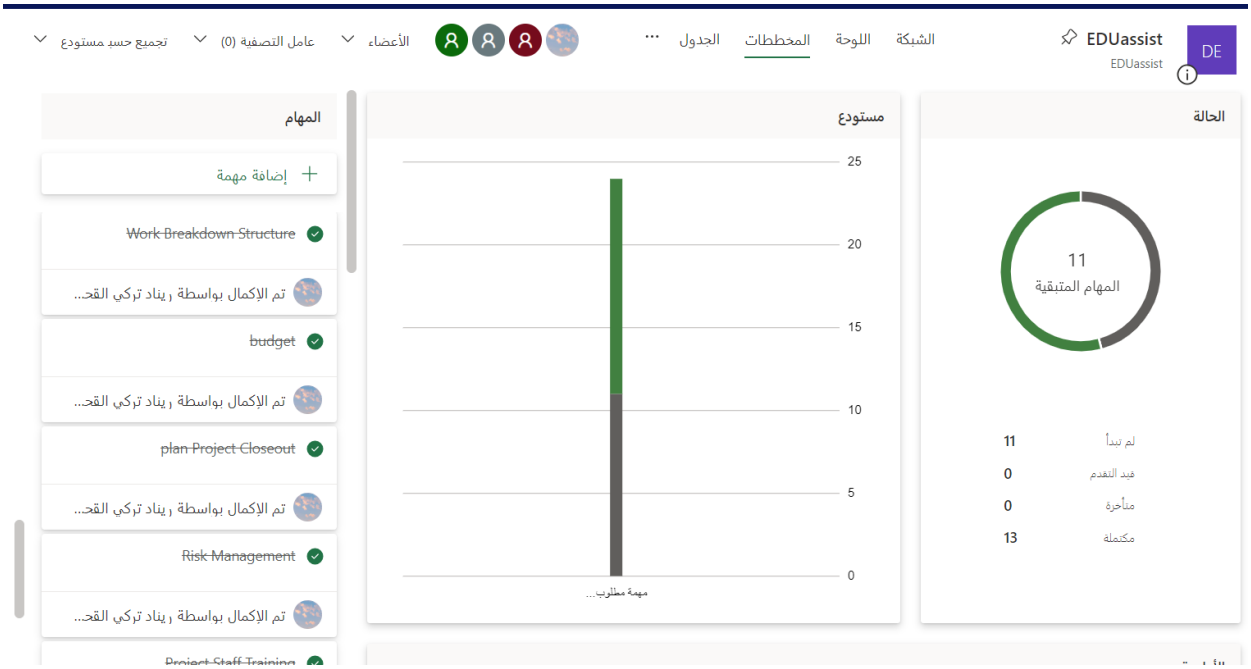


Figure 6: Project Schedule Chart

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3.2.3 Resource Allocation

This resource allocation plan provides essential support for each work activity, promoting efficient execution and enhancing collaboration among team members throughout the project lifecycle

#	Work Activities	Resources Allocated
1	Project Definition	Personnel: All Team Members Administrative Support: D. E. Elshahawy Computing Resources: Laptop/iPad , Software Tools: Microsoft Office (Word)
2	Project Proposal	Personnel: All Team Members Administrative Support: D. E. Elshahawy Computing Resources: Laptop/iPad Software Tools: Microsoft Office (One Drive), Microsoft Office (Word)
3	Project Management Plan (SPMP)	Personnel: All Team Members Administrative Support: D. E. Elshahawy Computing Resources: Laptop/iPad Software Tools: Microsoft Office (One Drive), Microsoft Office (Planner),canva, Notability, Zoom
4	Project Requirements	Personnel: All Team Members Administrative Support: D. E. Elshahawy Computing Resources: Laptop/iPad Software Tools: Microsoft Office (Word), Microsoft Office (One Drive),canva
5	Project Interface Design (SDS)	Personnel: All Team Members Administrative Support: D. E. Elshahawy Computing Resources: Laptop/iPad Software Tools: Microsoft Office (Word), Visual Studio Code, Figma
6	Project Test Plan (STP)	Personnel: 3 Members Computing Resources: 1 Desktop, 2 Laptops Software Tools: Selenium or Junit ,Microsoft Office (Word), Visual Studio Code, Figma , MySQL GitHub
7	Final Project	Personnel: 2 Members Computing Resources: 1 Desktop Software Tools: Microsoft Office (Word) Microsoft Office (PowerPoint) Visual Studio Code, Figma , canva Administrative Support: D. Enas Elsharawy

Table 9: Resource Allocation

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3.2.4 Budget Allocation

The EduAssist project will adopt a hybrid programming approach to develop an application that integrates AI functionalities. This strategy combines elements of both ground-up and existing system methodologies, which may extend the development timeline but allows for substantial cost savings and reduces errors by leveraging insights from previous systems without incurring additional expenses.

Given the focus on app development, the project will require the following resources:

- **Computing Resources:** We will need three computers and three to four laptops, with an estimated total cost of approximately 48,000 SR, as detailed in Table 4.
- **Software Tools:** Essential tools for building AI functionalities will be required, estimated at 6,000 SR each.
- **Special Testing and Simulation Facilities:** These will be necessary to ensure the application performs as intended.
- **Administrative Support:** Resources similar to those used by leading software companies, such as Google, will be needed to facilitate project management and operations.

In conclusion, the total budget for the EduAssist project is estimated at 280,000 SR. A comprehensive breakdown of the necessary resource allocations for each major work activity is provided in the WBS. Each resource type will be distinctly itemized within the activity budget to ensure clarity and facilitate effective financial management.

3.3 Project Tracking Plan

3.3.1 Requirements Management

The **Requirements Management** process for the EDUassist project is designed to rigorously manage, monitor, and control any changes to the project requirements, ensuring that they are aligned with project objectives and stakeholder expectations.

- **Change Control and Measurement:** All requirement changes will be subject to a formal **change control process**, which includes detailed documentation, evaluation, and reporting. A **Change Control Board (CCB)** will be established to review, assess, and authorize any proposed changes. The board will ensure that all modifications are communicated to relevant stakeholders, ensuring transparency and consensus.

- **Impact Assessment**: Each proposed requirement change will undergo a thorough **impact assessment** to evaluate its effects on **project scope**, **product quality**, **schedule**, **budget**, **resource allocation**, and **risk factors**. This analysis ensures that no change undermines the project's overall integrity. Only after the CCB reviews and approves the assessment will the change be implemented.

- **Configuration Management**: The **configuration management** process will include well-defined **change control procedures**. The **Change Control Board** will manage these procedures, ensuring that any modifications are reviewed, authorized, and integrated into the project in a controlled manner, with clear accountability.

- **Traceability and Reviews**: To maintain project integrity, we will implement **requirements traceability** to link all changes to specific project components. This

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approach ensures that every change is tracked from initiation to implementation. When necessary, **prototyping** and **modeling** will be utilized to validate changes before approval. Regular **impact analysis** and **review sessions** will be conducted to ensure that any changes are systematically evaluated and integrated without compromising project deliverables, timelines, or overall quality.

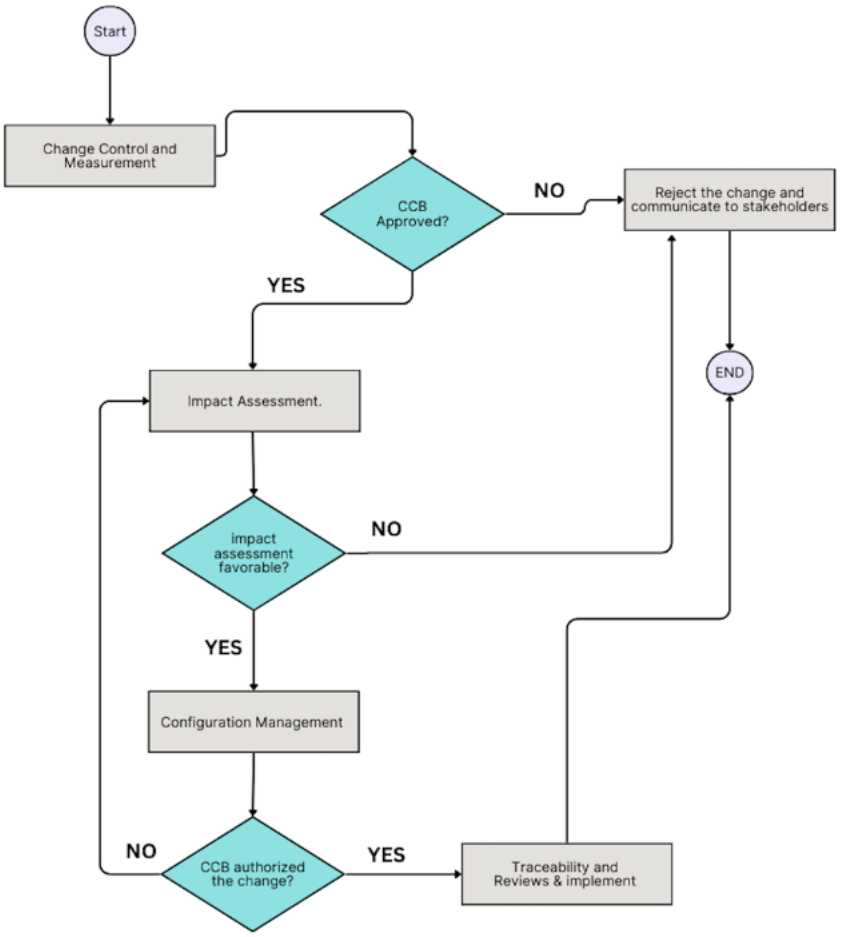


Figure 7: Requirement's Statue

3.3.2 Schedule Control

The schedule control for the EDUassist project will be carefully monitored to ensure that progress is consistently measured across all work activities, as detailed in Section 3.2.1. A variety of techniques will be employed to guarantee that actual progress aligns with the project plan. This includes regular reviews and the tracking of tasks at each project stage, helping to maintain the project timeline and avoid potential delays. Progress will be measured at both major and minor project milestones by tracking the completion of key tasks. Task boards will be used to visually organize and monitor the status of these tasks, allowing the team to clearly see which tasks are completed, in progress, or pending.To ensure the project stays on track, weekly meetings will be conducted to assess the progress of work and compare it to the planned schedule. The visual representation of tasks on the task board will help the team quickly identify any delays or bottlenecks in the workflow. If actual progress does not match the planned schedule, corrective measures will be taken. These may include reprioritizing tasks, reallocating resources, or extending work hours to

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meet deadlines. The goal is to quickly address any deviations to keep the project on course.

The project will utilize a task board, such as Microsoft Planner, to visually represent the project's progress. Tasks will be organized into categories such as "To Do," "In Progress," and "Done," making it easy for the team to track the status of each task at a glance. Each task will be assigned to specific team members with clear deadlines and priorities to ensure that responsibilities are well-defined. The team will regularly update the status of tasks, ensuring transparency and allowing everyone to remain informed of the project's overall progress.

Each project milestone will be evaluated based on the completion of key deliverables, which will be reflected as completed tasks on the task board. To ensure quality, team members will review and verify each other's work, ensuring that it meets the required standards before progressing to the next stage. Timeliness will also be a key metric, with the actual time taken to complete tasks compared against the planned schedule to determine whether work is being completed within the designated timeframe.

By following these processes and utilizing task boards for organizing and tracking tasks, the EDUassist project team will ensure that progress is consistently measured, quality is maintained, and any schedule deviations are promptly addressed.

3.3.3 Quality Control

The EDUassist project will implement several processes to ensure the quality of both the work being done and the deliverables produced. Quality control efforts will focus on ensuring that all tasks and outputs adhere to predefined standards, and that the final products meet the required specifications. These processes will be integrated into the project workflow to maintain consistency and accuracy across all stages of development.

All team members will adhere to predefined workflows to ensure consistency in tasks and deliverables. This includes compliance with established coding standards, documentation formats, and deadlines. Consistency in following these procedures will help maintain the quality of the project and ensure that each component is developed in line with project requirements.

Verification and Validation (V&V) processes will be implemented at key stages to guarantee that deliverables meet both the technical and user requirements. Verification will ensure that the deliverables conform to the project specifications, while validation will confirm that the products meet user needs and expectations. These checks will occur throughout the project lifecycle to catch potential issues early and ensure the final product is fit for purpose.

The project will also use regular peer and joint reviews as a core element of quality control. Team members will frequently review each other's work to identify defects and ensure compliance with project standards. These peer reviews will foster a collaborative environment where team members can provide feedback and address quality issues as they arise. Joint reviews will be conducted at major milestones to assess both the progress and quality of the deliverables, ensuring that the project stays on track and meets its goals.

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To further ensure compliance with project standards, periodic internal audits will be conducted. These audits will assess the team's adherence to processes and identify areas for improvement. By regularly evaluating compliance, the team can make adjustments to processes where necessary to improve efficiency and quality.

The use of quality control processes will be critical in maintaining high standards throughout the project. Quality Assurance (QA) activities will be implemented to ensure that work processes conform to the established project standards through regular monitoring. Additionally, verification and validation activities will verify that products meet the specified requirements and validate that they are functional and meet user expectations. Peer reviews, joint reviews, and internal audits will be used to continuously assess both the work processes and deliverables, ensuring that quality is upheld at every stage.

By following these quality control processes, the EDUassist project team will ensure that all deliverables meet the required specifications and that work is completed according to predefined standards. Regular checks, reviews, and audits will help to maintain quality throughout the project lifecycle, allowing the team to identify and address any issues promptly.

3.3.4 Reporting

The team will generate comprehensive reports at the conclusion of each project phase, which will be submitted to the project manager. These reports will include the software project proposal, Software Project Management Plan (SPMP), Software Requirements Specification (SRS), Software Design Specification (SDS), and Software Testing Plan (STP). The project manager will meticulously review and assess these documents to ensure effective oversight and alignment with project objectives.

Moreover, team members will leverage internal communication through social media platforms in conjunction with formal reporting to monitor project progress. This strategy allows for real-time updates and enhances the team's ability to address any potential misunderstandings swiftly. By integrating informal communication methods with official reports, the project can achieve greater efficiency and responsiveness.

3.3.5 Project Metrics

The EduAssist project will evaluate quality through time management, team collaboration, budget adherence, and productivity. A comprehensive project plan will be developed using a hybrid approach that combines agile and waterfall methodologies, segmenting the project into manageable components with defined timelines.To collect and retain metrics, we will employ tools such as Microsoft Project for planning and Slack for communication. Key metrics will focus on task completion time, budget utilization, and team performance. Data will be collected at the end of each phase and during weekly meetings, with validation processes ensuring accuracy.

Regular analysis of these metrics will inform trends and improvement areas, while findings will be reported through formal documentation and presentations. Accurate budget estimation and effective collaboration among team members will further enhance project success, positioning EduAssist to meet its objectives efficiently.

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3.4 Risk Management Plan

Risk management will involve several key activities throughout the project. As shown in **Figure 8**, the four main stages of risk management are risk identification, risk analysis, risk planning, and risk monitoring. Ongoing risk identification and assessment will be conducted at various stages of the project through regular risk assessment workshops. The team will also engage in risk mitigation work as part of their ongoing tasks. Contingency plans will be developed for high-priority risks to ensure alternative actions are available if needed. Additionally, **Table 10** outlines the potential risks associated with the project—such as technological, scheduling, and personnel risks—and the corresponding actions to address them, ensuring a clear plan is in place for managing each risk.

The risk manager will oversee these activities, ensuring that risks are regularly documented and tracked in the risk register, with updates being made as risks evolve. This includes the evaluation of changes in risk severity and likelihood, and adjustments to mitigation plans where necessary. Regular risk reports will be prepared, and risk communication procedures will be followed to ensure that significant updates or changes are promptly shared with all relevant stakeholders throughout the project lifecycle.



Figure 8: Risk Management Process

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#	Risk	Type of Risk	Probability	Impact	Actions for Mitigation	Avoidance Strategy
1	Changing customer requirements	Requirement	Low	Tolerable	Identify obstacles and repercussions from changes. Plan for clear, fixed requirements early.	Clearly define requirements and establish firm standards during the planning phase.
2	Financial constraints affecting accessibility of software tools	Technology	High	Tolerable	Explore alternative tools of equivalent quality.	Allocate a sufficient budget and create backup plans for procurement.
3	Economic challenges affecting budget allocation	Organizational	Low	Catastrophic	Review costs and budgets from the early stages.	Establish clear budget parameters at the beginning of the project.
4	Team members' well-being affecting project progress	Project	Moderate	Serious	Adjust schedules and timeframes to fit the current progress.	Develop a detailed schedule and use mobile reminders to track deadlines.
5	Failing to meet designated deadlines	Project	Low	Serious	Adjust schedules and timeframes to fit the current progress.	Develop a detailed schedule and use mobile reminders to track deadlines.
6	Insufficient experience among team members	Project, People	Moderate	Serious	Ensure regular communication between the project manager and the overseer.	Review project details early, and clarify uncertainties with supervisors.

Table 10: Potential Risks and Avoidanc

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3.5 Project Closeout Plan

This section outlines the project status and delivery methods for the EduAssist project. Several key stages have been completed, as illustrated in **Figure 9**. Throughout the project, team members have enhanced their skills and ventured into new areas of expertise. Each task is assigned to two individuals: one with expertise in that area and another gaining practical experience. This strategy minimizes the risk of reliance on any single team member.

This collaborative approach fosters an environment of shared learning and growth, enabling team members to enhance their skills and discover their potential. The closeout phase will involve a thorough review of project outcomes, lessons learned, and recommendations for future initiatives, ensuring that these insights support the ongoing development of the team and the success of future projects.

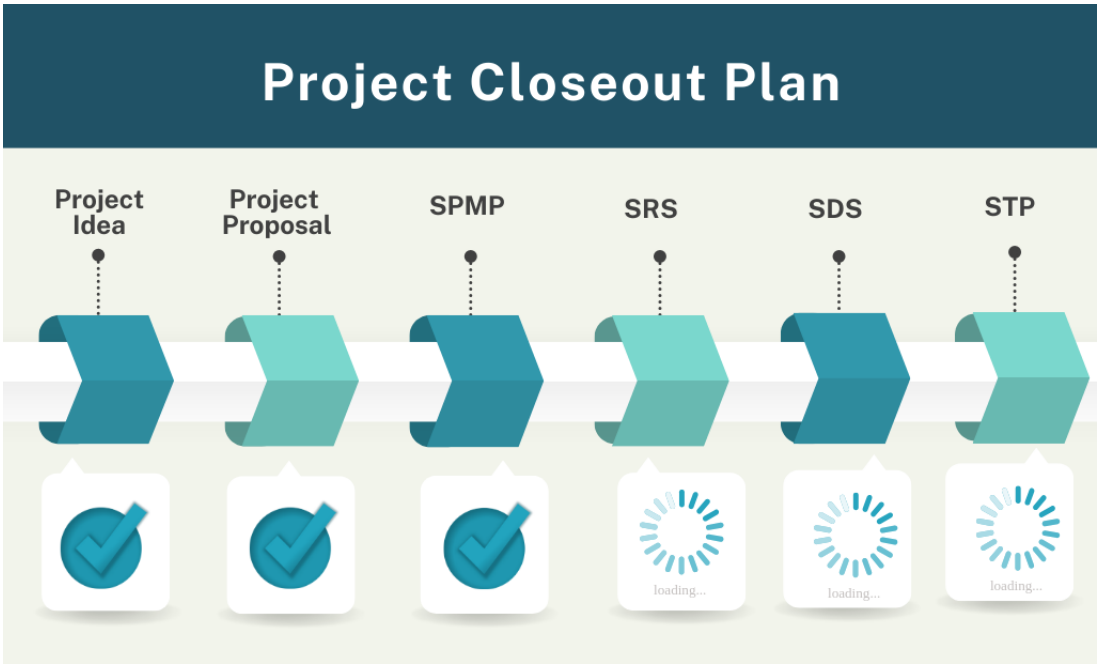


Figure 9: Project stage's status

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4. Technical Process Plans

This section of the Technical Process Plans provides a comprehensive overview of the process model for the EDUassist project. The plan outlines the key activities, relationships, and processes involved in the development of the application. Additionally, it includes a framework for managing the final deliverables.

4.1 Process Model

The EDUassist project will implement a Waterfall methodology, which is a linear and sequential approach to software development. This model emphasizes clear phases where each stage must be completed before moving on to the next. The relationships among the major project work activities are detailed in Table

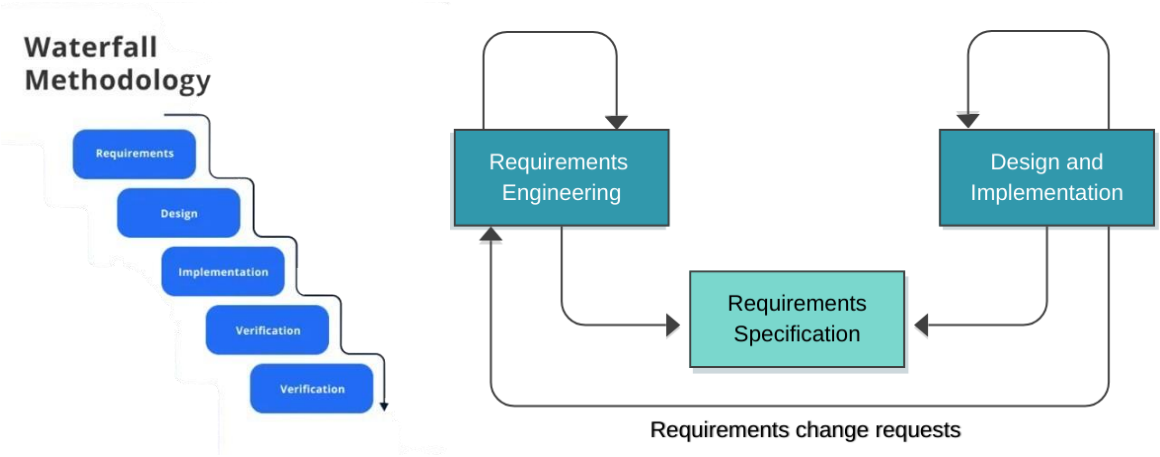


Figure 10: Waterfall Methodology and Plan-driven Model Process Phases.

Phase	Description
Requirements Engineering	Initial gathering of user needs and defining project scope.
Requirements Specification	Documenting the requirements and constraints that guide the project.
Design and Implementation	Creating design specifications and developing the application.
Testing	Systematic verification and validation of the application against requirements.
Project Closure	Finalizing documentation, conducting reviews, and evaluating project outcomes.

Table 11: Description of Plan-driven Model Process Phases

The relationships among these phases are linear, as illustrated in the diagram below. This structured approach ensures that each phase is completed thoroughly before progressing to the next, minimizing the risk of project scope creep.

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4.2 Methods, Tools, and Techniques

The methods, tools, and techniques for the EDUassist project are selected to enhance the development process and ensure successful project delivery. The following table summarizes the phases, associated tools and techniques, and relevant technical standards.

Phase	Tools / Techniques	Methods	Technical Standards
Requirements Engineering	MacOS Microsoft Windows 10 Microsoft Office 365 Physical Meeting Canva Figma Zoom Meetings GitHub	Table Diagrams	IEEE Standards 1058 - 1998
Requirements Specification	MacOS Microsoft Windows 10 Microsoft Office 365 Physical Meeting Zoom Meetings Canva Figma Miro OneDrive	Table Diagrams	IEEE Standards 1058 - 1998
Design and Implementation	Microsoft Office (Word) Visual Studio Code Figma MySQL GitHub	UML ER Diagram GUI Design Structure Design Text Design	IEEE Standards 1058 - 1998

Table 12 : Method, tools, and techniques standards

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4.3 Infrastructure

Software development for the EDUassist project necessitates a comprehensive plan for establishing and maintaining the development environment. This includes considerations for hardware, operating systems, networks, software tools, and various support facilities. The infrastructure details are outlined in the following table:

Software Development Infrastructure	
Hardware	Each team member is required to possess an individual workstation with adequate specifications for software development.
Network	Every team member must have access to a secure local area network to facilitate collaboration and data sharing.
Software	The development team will utilize tools such as Microsoft 365 for project management, Figma for design, and AWS for cloud services.
Policies	- All members are accountable for finishing their assigned tasks before the deadlines. - Attendance at team meetings is required for all members.
Facilities	All team members work in the designated office space at the institution, ensuring access to necessary resources and support services.

Table 13 : Infrastructure Plan

4.4 Product Acceptance

The primary goal of the EDUassist project is to fulfill the needs of educational institutions and enhance the learning experience for students. The project supervisor, Dr. Enas Al-Sharqawi, will review each deliverable, after which the proposed solutions will be adopted by the relevant stakeholders following these stages. Specify objective criteria for determining acceptability of the deliverables.

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5. Supporting Process Plans

This section outlines the documentation for the EDUassist software project.

5.1 Documentation

A range of documents is required to be submitted at various milestones throughout the project. The documents pertinent to the EDUassist project are detailed in the following table

Documents	Deadline	Format Standard	Prepare Document by Team Member	Review Documents
Defining Project	25-Aug-2024	Provided by the Project Supervisor	All the team members	Project Supervisor: Dr, Enas Elsharawy
Project Proposal	15-Sep-2024			
Project Management Plan (SPMP)	06-Oct-2024	EEE Std 1058-1998		
Project Requirement (SRS)	27-Oct-2024	IEEE 830-1998		
Project Design (SDS)	3-Nov-2024	IEEE Std 1016 – 1998		
Project Test Plan (STP)	1-Dec-2024	IEEE Std 829 – 1998		

Table 14: Documentation

6.Additional Plans

The additional plan comprises strategies aimed at enhancing the EDUassist project. One key initiative involves integrating collaborative features to boost student engagement and foster peer interaction. For instance, implementing a virtual whiteboard for group brainstorming sessions can facilitate real-time collaboration among students, making group projects more interactive. Additionally, personalized study recommendations based on AI analysis of individual learning styles will be introduced, helping students optimize their study routines. Another exciting enhancement is the incorporation of gamified learning elements, which can make academic tasks more enjoyable and motivate students to achieve their goals. All team members are dedicated to making informed decisions that will ensure EDUassist effectively supports students in their academic journeys.

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